Classical Field Theory — Exercise Sheet 13

Radiation estimates and boundary conditions

Solutions

Use Lecture 13's real Maxwell fields, retarded power theorem and amplitude $\mathcal{A}(u,n) = (4\pi)^{-1}\int \mathbf{J}(u+n\cdot y, y)\mathrm{d}^3y$. Here $n\in\mathbb{S}^2$, $u=t-r$, and $P_{\mathrm{T}}(n)=I-nn^{\mathsf{T}}$. The third problem also uses Lecture 6's time-reflected advanced kernel. The singular-source formula in the last problem is supplied separately.

Exercise 13.1 (A general slow-time source). Let $\rho_0(s,y)$ be scalar and $J_0(s,y)$ vector-valued, both real and smooth with compact spacetime support, satisfying $\partial_s\rho_0 + \mathrm{div}_y J_0 = 0$. For $0<\varepsilon\le 1$ put

$$\rho_\varepsilon(t,y)=\rho_0(\varepsilon t,y),\qquad \mathbf{J}_\varepsilon(t,y)=\varepsilon J_0(\varepsilon t,y),\qquad p_0(s)=\int y\rho_0(s,y)\mathrm{d}^3y.$$

Define finite constants

$$M_0=\int \sup_s|\partial_s J_0(s,y)|\mathrm{d}^3y,$$
$$M_1=\int |y|\sup_s|\partial_s^2 J_0(s,y)|\mathrm{d}^3y,$$
$$M_2=\frac12\int |y|^2\sup_s|\partial_s^3 J_0(s,y)|\mathrm{d}^3y.$$

(a) Verify conservation and prove $p_0'(s)=\int J_0(s,y)\mathrm{d}^3y$. Find the scaled dipole $p_\varepsilon(t)=\int y\rho_\varepsilon(t,y)\mathrm{d}^3y$ and its second derivative.

(b) Prove, uniformly for all $u\in\mathbb{R}$, the explicit absolute estimate

$$\left|P_\varepsilon(u)-\frac{|\ddot p_\varepsilon(u)|^2}{6\pi}\right|\le\frac{2M_0M_1+M_1^2}{4\pi}\varepsilon^5.$$

(c) Improve the integrated error to $O(\varepsilon^6)$ by retaining the first delayed-sampling correction. Give an explicit bound using M_0,M_1,M_2. Why does this improvement not assert that the directional power density has an $O(\varepsilon^6)$ error?

Solution. For the first part, both terms of the conservation law acquire the same factor ε. Compact spatial support permits coordinate integration by parts, giving

$$p_{0,i}'=-\int y_i\partial_{y_j}J_{0,j}\mathrm{d}^3y=\int J_{0,i}\mathrm{d}^3y,\qquad p_\varepsilon(t)=p_0(\varepsilon t),\qquad \ddot p_\varepsilon(t)=\varepsilon^2\int\partial_s J_0(\varepsilon t,y)\mathrm{d}^3y.$$

The scaled current is compact in spacetime for each positive ε, so Lecture 13 applies, although the temporal support grows as ε decreases.

For the second part, set $s=\varepsilon u$ and $a(s)=\int\partial_s J_0(s,y)\mathrm{d}^3y$. Differentiation of the amplitude and the mean-value theorem give

$$\mathcal{A}_\varepsilon'(u,n)=\frac{\varepsilon^2 a(s)}{4\pi}+R,\qquad |a(s)|\le M_0,\qquad |R|\le\frac{\varepsilon^3M_1}{4\pi}.$$

Since P_{T} is a contraction, the difference of the squared transverse norms is bounded by twice the product of these bounds plus $|R|^2$. Integrating over a sphere of area 4π yields

$$\left|P_\varepsilon-\frac{|\varepsilon^2 a|^2}{6\pi}\right|\le\frac{2\varepsilon^5M_0M_1+\varepsilon^6M_1^2}{4\pi}.$$

Here the leading integral uses $\int |n\times a|^2 \mathrm{d}\Omega = 8\pi|a|^2/3$. The dipole identity above and $\varepsilon \le 1$ prove the requested estimate.

For the final part, Taylor's formula with a bounded second derivative of $\partial_s J_0$ gives

$$4\pi\mathcal{A}'_\varepsilon = \varepsilon^2 a(s) + \varepsilon^3 m(s,n) + \varepsilon^4 b_\varepsilon(s,n), \quad m = \int (n\cdot y)\partial_s^2 J_0(s,y)\mathrm{d}^3y, \quad |m| \le M_1, \quad |b_\varepsilon| \le M_2.$$

The vector $m(s,-n) = -m(s,n)$, while $P_{\mathrm{T}}(-n) = P_{\mathrm{T}}(n)$. Thus $\int P_{\mathrm{T}}a \cdot P_{\mathrm{T}}m\mathrm{d}\Omega = 0$. The remaining cross terms and squares are bounded by

$$\left| P_\varepsilon - \frac{|\ddot{p}_\varepsilon|^2}{6\pi} \right| \le \frac{\varepsilon^6}{4\pi}\left(2M_0M_2 + M_1^2 + 2M_1M_2 + M_2^2\right).$$

All constants are independent of u and ε. At a fixed direction the first cross term generally remains nonzero; its cancellation is a property of the integrated power.

Exercise 13.2 (A nonradial source at zero acceleration). Choose a nonnegative $f \in C_c^\infty(B_1(0))$ with $\int f = 1$ and nonzero first moment $\mu = \int y f(y)\mathrm{d}^3y$. Let $a \in \mathbb{R}^3 \setminus \{0\}$ and choose a smooth compactly supported p_0 equal to as^3 on $|s| \le 1$. Use the conserved dipole source

$$p(t) = p_0(\varepsilon t), \qquad \rho = -p(t)\cdot\nabla f, \qquad \mathbf{J} = \dot{p}(t)f, \qquad 0 < \varepsilon \le 1.$$

Compute the exact retarded power $P(0)$. Show that it is positive although $\ddot{p}(0) = 0$, and evaluate it for $a = e_3$, $\mu = e_3/4$. Explain what this rules out about a relative error estimate. You may use the angular moments

$$\int n_i n_j \mathrm{d}\Omega = \frac{4\pi}{3}\delta_{ij}, \qquad \int n_i n_j n_k n_l \mathrm{d}\Omega = \frac{4\pi}{15}(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Solution. The sampled acceleration is exactly $\ddot{p}(n\cdot y) = 6\varepsilon^3(n\cdot y)a$. The amplitude and power therefore are

$$\mathcal{A}'(0,n) = \frac{6\varepsilon^3}{4\pi}(n\cdot\mu)a, \qquad P(0) = \frac{36\varepsilon^6}{16\pi^2}\int (n\cdot\mu)^2\left(|a|^2 - (n\cdot a)^2\right)\mathrm{d}\Omega.$$

Contracting the supplied second and fourth moments gives

$$\int (n\cdot\mu)^2|a|^2\mathrm{d}\Omega = \frac{4\pi}{3}|a|^2|\mu|^2, \qquad \int (n\cdot\mu)^2(n\cdot a)^2\mathrm{d}\Omega = \frac{4\pi}{15}\left(|a|^2|\mu|^2 + 2(a\cdot\mu)^2\right).$$

Their difference yields

$$P(0) = \frac{6\varepsilon^6}{5\pi}\left(2|a|^2|\mu|^2 - (a\cdot\mu)^2\right).$$

Cauchy–Schwarz bounds its parenthesis below by $|a|^2|\mu|^2 > 0$. For $a = e_3$ and $\mu = e_3/4$, that parenthesis is $1/16$, giving $P(0) = 3\varepsilon^6/(40\pi)$. Such an f exists: translate a normalized radial smooth bump of radius less than $1/4$ to centre $e_3/4$. But the instantaneous leading dipole power is zero. An estimate of the error bounded by a finite constant times that leading value would force $P(0) = 0$, contradicting the exact result. The absolute estimates of the first problem allow it.

Exercise 13.3 (The half-retarded, half-advanced field). For a real smooth compact spacetime conserved source, let $(\phi_{\mathrm{em}}^{\mathrm{ret}}, \mathbf{A}_{\mathrm{ret}})$ be its retarded potentials. Define the advanced potentials by replacing $t - |x-y|$ by $t + |x-y|$ in the same $1/(4\pi|x-y|)$ integrals. Time reflection of Lecture 6's kernel gives these smooth sourced solutions. Let the symmetric potentials be half the sum of the two, and form their Maxwell fields.

(a) At future null infinity, $t = r + u$, show that the advanced field is exactly zero for all sufficiently large r (at each fixed u). Find the symmetric field's limiting outward power in terms of the retarded power.

(b) At past null infinity, $t = v - r$, define

$$\overline{\mathcal{A}}(v,n) = \frac{1}{4\pi}\int \mathbf{J}(v - n\cdot y, y)\mathrm{d}^3y, \qquad \overline{\mathcal{V}}(v,n) = \frac{1}{4\pi}\int \rho(v - n\cdot y, y)\mathrm{d}^3y.$$

Derive the leading advanced electric and magnetic coefficients and the signed outward power of the symmetric field. Primes now mean ∂_v. Does averaging the potentials cancel radiation at future null infinity?

Solution. For the first part, enclose the spatial support in B_R and the temporal support in $[-T,T]$. At $t = r + u$ the advanced source time is at least $2r + u - R$. Once this exceeds T, the source and all its derivatives vanish throughout the potential integrals. Hence the advanced potentials and fields are zero there. The symmetric field is exactly half the retarded field; the quadratic flux is one quarter of its retarded value.

For the second part, at $t = v - r$ the retarded source time is at most $v - 2r + R$, so the retarded fields vanish for large r. Expanding the advanced distance as in Lecture 13 gives

$$\mathbf{A}_{\mathrm{adv}} = \frac{\overline{\mathcal{A}}(v,n)}{r} + O(r^{-2}), \qquad \phi_{\mathrm{em}}^{\mathrm{adv}} = \frac{\overline{\mathcal{V}}(v,n)}{r} + O(r^{-2}).$$

Differentiating the exact source integrals at fixed t gives the leading Cartesian derivative $+n_i\overline{\mathcal{V}}'/r$ for a scalar potential; angular derivatives contribute only $O(r^{-2})$, by the same bounded-support estimates as in Lecture 13. The total spatial divergence of $\mathbf{J}(v - n\cdot y, y)$ integrates to zero and equals the integral of $\mathrm{div}_y\,\mathbf{J} - n\cdot\mathbf{J}_t$. Conservation therefore yields

$$\overline{\mathcal{V}}' = -n\cdot\overline{\mathcal{A}}', \qquad E_* = -(n\overline{\mathcal{V}}' + \overline{\mathcal{A}}') = -P_{\mathrm{T}}\overline{\mathcal{A}}', \qquad B_* = n\times\overline{\mathcal{A}}' = -n\times E_*.$$

Consequently $(E_*\times B_*)\cdot n = -|E_*|^2$. The symmetric field has the limiting signed outward power

$$-\frac{1}{4}\int_{\mathbb{S}^2} |n\times\overline{\mathcal{A}}'(v,n)|^2\mathrm{d}\Omega.$$

It transports energy inward at past infinity and outward at future infinity. These are different asymptotic regions, so the two fields do not cancel there. No pointwise equality between the past and future power functions has been assumed.

Exercise 13.4 (Liénard–Wiechert: a supplied singular-source formula). Let $z\colon \mathbb{R} \to \mathbb{R}^3$ be a smooth complete worldline with $|\dot{z}(s)| \le v_* < 1$. Fix an event (t, x) with $x \neq z(t)$. Let $Q \in \mathbb{R}$ be its charge. For this problem only, take as given the point-source potential prescription

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi}\int \frac{\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s, \qquad \mathbf{A}(t,x) = \frac{Q}{4\pi}\int \frac{\dot{z}(s)\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s.$$

The delta integrals are understood locally at their roots. You may use the one-variable identity $\delta(g(s)) = \sum_j \delta(s-s_j)/|g'(s_j)|$ for simple roots.

(a) Prove existence and uniqueness of the retarded root $t = s_{\mathrm{ret}} + |x - z(s_{\mathrm{ret}})|$. Evaluate the two potentials in terms of $R = |x - z(s_{\mathrm{ret}})|$, $n = (x - z(s_{\mathrm{ret}}))/R$ and $v = \dot{z}(s_{\mathrm{ret}})$.

(b) For $z(s) = (3s/5, 0, 0)$ at $(t, x) = (1, 0)$, compute the retarded time, Jacobian and scalar potential. Explain exactly which smooth-source hypotheses fail and why this calculation establishes no self-force law.

Solution. For the first part, put $h(s) = s + |x - z(s)|$. The speed bound gives, whenever $s_2 > s_1$,

$$h(s_2) - h(s_1) \ge (1 - v_*)(s_2 - s_1) > 0.$$

Also $h(s) \to +\infty$ as $s \to +\infty$, since $h(s) \ge s$. For negative s, $h(s) \le (1-v_*)s + |x - z(0)| \to -\infty$. Continuity and strict monotonicity give exactly one root $h(s) = t$. If its distance vanished, it would have $s = t$ and $x = z(t)$, excluded by the hypothesis. Thus $R > 0$ at the root and differentiation there is valid. For $g(s) = t - h(s)$ it gives

$$g'(s_{\mathrm{ret}}) = -(1 - n \cdot v), \qquad 1 - n \cdot v \ge 1 - v_* > 0.$$

The supplied delta identity now evaluates the prescriptions as

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi R(1 - n \cdot v)}, \qquad \mathbf{A}(t,x) = \frac{Qv}{4\pi R(1 - n \cdot v)}.$$

For the second part, $s = 5/8$ gives $R = 3/8$, $n = -e_1$ and $v = 3e_1/5$, hence $s + R = 1$ and $1 - n \cdot v = 8/5$. Substitution yields $\phi_{\mathrm{em}} = 5Q/(12\pi)$ and $\mathbf{A} = Qe_1/(4\pi)$. The associated current has a spatial point delta and persists for all time; it is neither a smooth spacetime source nor compactly supported in time. Thus this is an application of the explicitly supplied singular-source prescription, not an application of Lecture 13's smooth-source theorem. The fields are evaluated off the worldline. No finite value of the field on the charge, and no equation for its self-force, has been defined.